

Rare Earth Elements in China: Deposits, Production, and Policy

Executive Summary: China dominates global rare earth element (REE) supply through diverse deposit types and integrated processing. Major Chinese REE deposits include *ion-adsorption clays* in the South (e.g. Jiangxi, Guangdong, Fujian, Guangxi), and hard-rock deposits such as bastnäsite-bearing carbonatites (notably the Bayan Obo Fe-REE-Nb deposit in Inner Mongolia) and monazite/xenotime-rich sands and granites (e.g. southern Jiangxi, Shandong). China holds an estimated **44–50 million tonnes** REO of reserves [54†L131-L137] [33†L277-L284], ~50% of global known reserves. 83% of China's REE reserves lie in a single basin (Bayan Obo, ~36 Mt REO) [33†L296-L304]. China's annual mine output was roughly **210–240 kt REO** in recent years (~68–69% global production) [33†L283-L292] [35†L134-L139], with recent quotas around 255,000 t REO (2024). The industry is highly consolidated (a few state groups control >80% output) under tight regulation. Typical processing uses **flotation and magnetic separation** to concentrate hard-rock ores, followed by solvent-extraction or ion-exchange to separate individual REEs. Ion-adsorption ores are mined by in-situ or heap leaching (often using NH_4^+ or H_2SO_4), achieving recoveries of ~70–90% [9†L38-L46]. Environmental impacts include acidified tailings, water contamination, and radioactive residues from U/Th in minerals. China's *Ministry of Natural Resources* (MNR) and *Ministry of Ecology and Environment* (MEE) enforce strict quotas, export licenses and the new Rare Earths Resource Protection Law (2021) to control mining, usage and trade. Export of REE ores/oxides is heavily restricted; recent years saw bans (e.g. on exports to Russia) and multi-layer controls rather than outright bans [47†L39-L48] [54†L145-L150]. Supply chains remain fragile due to China's dominance, prompting global efforts to diversify. **Data gaps** include uncertainties in illegal production and unreported stocks; official Chinese data (e.g. yearbooks, MNR reports) and company figures are often opaque.

Major REE Deposit Types and Distribution

China's REE resources occur in both light-REE (LREE) and heavy-REE (HREE) mineral deposits, broadly classified as follows:

- **Ion-Adsorption Clays (Regolith-Hosted):** Found mainly in *Southern China* (Jiangxi, Guangdong, Fujian, Guangxi, Hunan, Guizhou, Yunnan). These weathered granitic profiles host REE adsorbed onto kaolinite/illite clays. They typically contain *mixed LREE-HREE* with higher HREE fractions than most global deposits, and are easily leachable. Geologically, they form on **Cretaceous–Paleogene** granite or pegmatite complexes, with ages ~100–60 Ma (e.g. Nanling and Yanshanian intrusives). Ironstone or gossan layers cap some deposits. The REE are present as exchangeable cations, chiefly associated with kaolinite (illite) surfaces [9†L39-L47]. These deposits host much of China's HREE; for example, Ganzhou (southern Jiangxi) and Longnan (western Jiangxi) yield >50% of China's HREE output [36†L42-L45].
- **Carbonatite-Hosted (Bastnäsite, Monazite, Parisite):** The world's largest REE deposit, **Bayan Obo** (Inner Mongolia), is an iron-REE-Nb orebearing carbonatite complex (Mesoproterozoic, ~1.2–1.3 Ga host rock ages) with stratiform bodies in dolomite/quartzite. Ores carry *bastnäsite*-(Ce) and *monazite*-(Ce) (LREE phosphates/carbonates), plus subordinate néodyme minerals. Bayan Obo holds >35 Mt REO [35†L139-L146] (about 83% of China's reserves), with grades ~3–5% REO

【17†L334-L342】. Other carbonatites (e.g. Maoniuping, Sichuan; Kangdian, Yunnan) and associated fenitization zones also supply LREEs. These deposits often contain scant uranium (<0.1 ppm) but moderate thorium (~0.3–1.3 wt% in monazite) 【17†L351-L354】.

- **Alkaline Igneous / Metasomatic (Loparite/Violarite, etc):** Small REE deposits occur in A-type granites and syenites in Northeast and East China, but are generally minor. Some older (e.g. Paleozoic) carbonatite-like fenites in Liaoning and Hubei host xenotime, monazite.
- **Placer (Heavy Mineral Sands):** Coastal sands in provinces like Shandong and Hebei contain *monazite* and *xenotime* as heavy minerals. These are minor relative to clays and bastnäsite but important sources of HREE and Th.
- **Lateritic Nickel-Cobalt Deposits:** REE (mainly Ce, La) in Ni-Co laterite ores (e.g. Gansu, Guangxi) are of industrial interest, though China's laterite REE potential is smaller than those of Indonesia/Philippines.

Table 1. China's major REE deposit types and characteristics.

Deposit Type	Main Locations (provinces)	Host Lithology / Age	REE Minerals	Features
Ion-adsorption clays	Southern China (Jiangxi, Guangdong, Guangxi, Fujian, Hunan, Guizhou, Yunnan)	Weathered Cretaceous–Paleogene granites (up to ~120 Ma)	Clays (kaolinite, illite); adsorbed REE cations (mostly Ce, La, Nd; significant Y, Dy)	Nearly 100% exchangeable REE; no major U/Th; ~low mining cost via in-situ leach 【9†L39-L47】
Carbonatite/bastnäsite	Inner Mongolia (Bayan Obo); Sichuan; Sichuan–Gansu border	Carbonatites (Mesoproterozoic intrusions ~1.2–1.3 Ga)	Bastnäsite (Ce, La fluorocarbonates), monazite, parisite	High Ce/La (LREE) content; Baotou (Bayan Obo) is largest REE deposit (~40 Mt REE minerals) 【17†L334-L342】
Granite-related LREE/HREE	Hunan, Jiangxi; also Norilsk-type (magmatic sulfides)	Mesozoic granites, pegmatites	Monazite, xenotime, fluorocarbonates	Moderate REE; often associated with U/Th in monazite/xenotime

Deposit Type	Main Locations (provinces)	Host Lithology / Age	REE Minerals	Features
Placer sands (heavy minerals)	Shandong, Hebei, Fujian (beach/riverine)	Quaternary sands derived from granite/mineral sources	Monazite-(Ce), xenotime, zircon, ilmenite	Separated by gravity/spiral; significant Th in monazite; continuous but low grade
Ion-adsorption laterite (Ni-Co)	Guangxi, Shanxi (lateritic profiles)	Cretaceous basalts (weathered)	LREEs adsorbed on iron oxides	Minor; REEs are byproduct of Ni/Co mining

China's REE **geologic ages** span Proterozoic (Bayan Obo ~1.2 Ga), Mesozoic (granites ~120–60 Ma forming clays) to Cenozoic (placer sands). Southern ion-adsorption deposits are associated with *Yanshanian* (Jurassic–Cretaceous) granites. The host lithologies reflect major tectono-magmatic events (e.g. Mesozoic South China magmatism) and later weathering. Mineralogically, China's resources encompass nearly the full REE suite: clays yield balanced LREE+HREE mixes **【33†L39-L47】**, while Bayan Obo is LREE-rich.

Production, Reserves and Trade

China has by far the largest share of both REE reserves and production. According to USGS and Chinese sources, China's REE reserves were **44–50 million tonnes REO** (as of 2024–2026) **【54†L133-L137】** **【33†L277-L284】**, or about half of global reserves. Of this, ~36 Mt REO (83%) is in Bayan Obo alone **【33†L296-L304】**. Southern provinces (Jiangxi, Fujian, Guangdong, Guangxi, Yunnan, Guizhou) collectively hold ~10 Mt REO (mainly in clays) **【33†L307-L314】**. Western Sichuan holds ~5 Mt (carbonatite-associated) **【33†L307-L314】**.

Production: China's annual mined REO peaked at ~240,000 t in 2021 (~68% of world output) and was about 210,000 t in 2022 **【35†L133-L139】**. Quotas and consolidation have since stabilized output (2024 quota ~255,000 t). Province-level output is concentrated in Inner Mongolia (Bayan Obo ~50% of Chinese output) and Sichuan (~12%), with the remainder largely from southern clays and other deposits **【36†L67-L70】** **【36†L62-L66】**. For example, 2021 figures (estimated) are: Northern (Inner Mongolia, Liaoning) ~120,000 t (mostly bastnäsite/monazite concentrate) and Southern (clays in Jiangxi/Fujian) providing heavy REO ~18,000 t **【36†L42-L45】** **【36†L55-L58】** (Jiangxi ~8,500 t, Fujian ~3,000 t). Light RE (La, Ce) from northern ores was ~168,000 t in 2019 **【36†L67-L70】**.

Reserves & Resources: According to USGS (2026), global REE reserves are ~91 Mt REO, with China ~48% (~44 Mt) **【54†L133-L137】**. A Chinese Academy of Sciences report estimates ~50 Mt **【33†L277-L284】**. By deposit, Bayan Obo (LREE mineral) has ~35 Mt **【35†L139-L146】**; ion-adsorption clays are smaller individually but numerous. Resource estimates often exclude low-grade or unmined potentials; illegal mining adds uncertainty.

Trade and Export: China is the dominant exporter of REE products (oxides, metals, magnets). Recent data: ~49,400 t REO exports in 2022, rising to ~55,000 t in 2023 **【35†L116-L124】**. Key importers of Chinese RE materials include Japan (70% of its needs in 2022 ~25,000 t) **【35†L116-L124】**, the EU, and

South Korea. Under China's multi-layer control, exports require MIIT licensing. Export quotas (historically set by state from 2006 onward) were largely replaced after WTO rulings, though tight administration persists 【47†L54-L62】 【47†L147-L155】. Recently (2023–2024) China imposed an effective ban on RE exports to Russia, affecting ~5,000 t/year 【35†L116-L124】. Despite quotas, significant volumes (16–25 kt/year) continue to flow legally to allies and allied companies under licensing.

Table 2. *China rare earth production trends.* (Estimates from USGS and industry sources)

Year	Chinese Mine Production (t REO)	Global Mine Production (t REO)	China's Share (%)
2015	~105,000 【52†L204-L212】	~105,000 【52†L204-L212】	~80%
2019	~210,000–220,000	~275,000	~78%
2021	~240,000 【36†L7-L10】	~350,000	~68%
2022	~210,000 【35†L134-L139】	~300,000	~70%
2023 (est)	~230,000	~330,000	~70%

Notes: Chinese figures include both bastnäsite/monazite concentrates and clay leach output. Global totals are rough and include non-Chinese producers (Australia, USA reopening, India, etc.). The dip in 2022 reflects quota limits; resurgence in 2023 reflects relaxation of controls and capacity additions in China.

Mining and Processing Methods

China uses deposit-specific mining and processing:

- **Mining Techniques:**

- *Clays (Ion-adsorption):* Typically mined by open-pit or shallow quarry; waste overburden removed to expose clay layers. Many operations use **in-situ leaching**: acid (H_2SO_4) or ammonium salt solution is pumped through drilled holes (injectors), leaching REE into solution which is pumped out 【9†L49-L52】. Heap leaching (stacking ore and spraying solution) and tank leaching are also used (legacy methods). Recovery rates of leached REEs are reported at ~70–90% 【9†L49-L52】.
- *Bastnäsite/Monazite ores:* Mined by open-pit (e.g. Bayan Obo) or underground methods, depending on deposit geometry. Ore is crushed and ground. Because bastnäsite and monazite are moderately magnetic and dense, the typical beneficiation involves gravity separation (spirals, shaking tables) and *magnetic separation* (low- and high-intensity) to upgrade the ore. **Flotation** is widely used in bastnäsite deposits to produce a mixed RE carbonate/phosphate concentrate (e.g. bastnäsite, parisite). Recovery from flotation often exceeds 85%. For instance, froth flotation of bastnäsite can yield >80% REE recovery at ~50% concentrate grade 【14†L203-L211】 (bastnäsite is analogous to Mountain Pass ore).
- *Monazite sands:* Beach/mining sands are dredged or trenched, then subjected to wet gravity separation (spirals) to isolate heavy minerals including monazite and xenotime. *Electrostatic separation* and magnetic sorting further concentrate these.
- *Laterites:* Nickel-cobalt laterite mines may produce a REE-enriched tail or separate layer; some in China use acid leaching of limonite zones for REE as byproduct.

- **Processing (Beneficiation to Refined Products):**

China's processing is vertically integrated. After concentration, the main steps are roasting or acid digestion (for hard minerals) followed by solvent extraction (SX) to separate individual REEs. For *bastnäsite/monazite*, common routes include:

- *Sulfuric or hydrochloric acid leach* to dissolve REE-bearing carbonates and phosphates. For example, monazite is often digested with concentrated H₂SO₄ at high temperature.
- The leach solution (REE sulfates or chlorides) is then subjected to solvent extraction with organic extractants (e.g. acidic organophosphorus reagents like D2EHPA, Cyanex 272, TBP). China has industrialized SX processes that sequentially separate light vs. heavy REEs 【54†L131-L137】. Reports indicate global leadership: China's share of global REE separation capacity is ~85% 【35†L120-L124】.
- *Magnet separation*: Not for concentration of REE (after flotation), but magnetic alloys (NdFeB magnets) are made in China, where >90% of world magnet output is produced.

Recovery rates: End-to-end recovery (ore to oxide) is high for bastnäsite: typical REE recovery by SX is 85–95%, with final oxide purity often >99% for individual elements. Heavy-ion adsorption clays yield ~70–90% of contained REE 【9†L49-L52】, though environmental factors (heterogeneity) can lower efficiency.

- **Technology and Scale**: China has developed new methods such as *alkaline leaching*, *bioleaching*, and *membrane separation* in research. However, traditional SX dominates in industry. Chinese companies (e.g. Baotou, Ganzhou, Weiqiao) operate pilot projects on chromatographic or ion-exchange systems, but these supplement rather than replace SX.

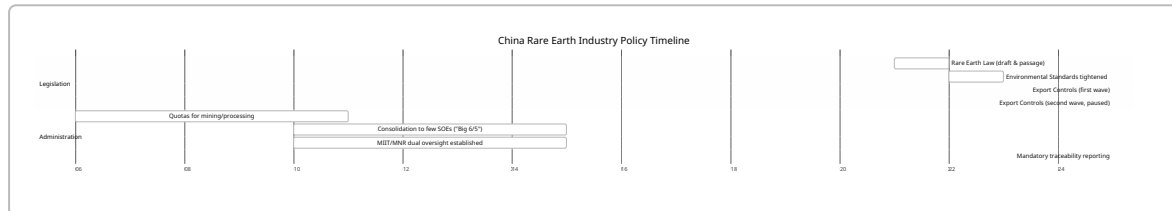
Environmental Impacts and Regulation

REE mining and processing have significant environmental challenges.

- **Waste & Pollution**: Extraction (especially heap/in-situ leaching) produces large volumes of tailings and waste rock. For every tonne of REE produced at Bayan Obo, an estimated 2,000 tonnes of waste gas (HF, SO₂, particulates) and 75 m³ of *acidic wastewater* are emitted 【17†L319-L327】. Tailings ponds (like Baotou's) accumulate acid-generating residues and radioactive elements (thorium, uranium in monazite). Groundwater contamination with REEs and fluorides has been documented near processing sites 【25†L53-L62】. Southern clay mines have issues of spent leachate containing ammonium or sulfate, which can alter soil and water chemistry.
- **Remediation**: China has begun large-scale revegetation of tailings and adoption of “green mining” (green lixiviants, closed-loop water use). For example, Baotou's tailings ponds are being afforested and stabilized 【24†L73-L82】. New studies focus on bioremediation (microbes/plants to uptake REEs) and neutralizing agents for effluents 【9†L49-L52】 【24†L66-L75】. Reprocessing of tailings is also explored to recover residual REEs and mitigate waste.
- **Radioactivity**: Many REE minerals carry U/Th (monazite ~3–10% ThO₂, xenotime ~2–3% ThO₂) 【17†L351-L354】. Chinese environmental regulations require controlled disposal or reuse of radioactive by-products. Large volumes of “radioactive” slag from early REE smelting have raised health concerns (as reported by BBC 【17†L341-L344】). Modern plants monitor and restrict Th/U releases; some by-products (e.g. Th) are recycled for nuclear fuel or industry.
- **Regulatory Framework**: The **Ministry of Natural Resources (MNR)** oversees resource evaluation, allocation of mining rights, and quotas on mining and processing volume. The **Ministry of Ecology and Environment (MEE)** enforces pollution controls and mining permits. Key laws include the revised *Mineral Resources Law (2020)* and the *Rare Earth Resources Protection Law (2021)*, which aim to prevent illegal mining and enforce restoration. Environmental Impact

Assessments are mandatory for new mines, per national standards. Export licensing now requires demonstration of legal provenance and environmental compliance.

Recent policies have tightened oversight. In July 2025 China enacted new interim rules requiring monthly tracking of production against quotas and stricter record-keeping 【33†L232-L240】. These aim to curb fraud and unsanctioned output. Overall, China's regulatory regime now spans mining, processing and trade (see timeline below).



(Mermaid timeline illustrating key rare earth policies. "Big 6/5" refers to the 2010s consolidation of the industry into a handful of state groups.)

Supply-Chain and Geopolitical Implications

China's rare-earth strategy is pivotal to global tech supply chains. China accounted for **60–70%** of world mine output in 2021–23 【35†L133-L139】 and **85% of separation capacity** 【35†L120-L124】. It also produces ~92% of NdFeB permanent magnets 【35†L124-L127】. This concentration gives China leverage: it has used export restrictions (e.g. 2010–11 quotas, 2023 Russia ban) as strategic tools. However, actual controls are multi-layered. According to analysts, China's regime is not a simple "export ban" but a system of domestic quotas (administered by MIIT) and selective export licensing (with MNR tracking origin) 【47†L39-L48】 【47†L147-L155】. Only government-approved mines and processors may legally produce, limiting supply nodes.

In response, consuming countries are diversifying (new projects in Australia, US, Southeast Asia; recycling initiatives). Japan and the EU have invested in rare-earth development overseas. The US has restarted its Mountain Pass (California) and opened a new HREE mine (Round Top, Texas). Nonetheless, China's dominance remains: one analysis projects the West will still rely on China for ~90% of its HREE needs by 2030 【53†L19-L23】.

Domestically, China is building stockpiles of light and heavy REOs (estimated strategic reserve on the order of tens of thousands of tons) to buffer supply. It also requires onshore end-use (e.g. banning export of Nd magnets in raw form) to capture added value. Internationally, rare earths are now entwined with trade/political conflicts: e.g. US-China trade tensions have seen both rare-earth shortages in downstream industries and renewed focus on secure supply chains.

Major Companies: The industry is dominated by state-linked firms: *China Northern Rare Earth Group* (Baotou, Bayan Obo) and *China Southern Rare Earth Group* (Ganzhou), among a few conglomerates (often part of CREEC/Minmetals). Other key players include *Chinalco Rare Earths*, *Xiamen Tungsten Co* (through PRC Tungsten group), and *Jiangxi Ganzhou Rare Earth*. Many are vertically integrated (mining through magnet production). A "Big Five" (now five groups) were designated around 2015 to concentrate resources and enforce policy.

Tables, Maps, and Figures

(Embedded images and figures to illustrate deposit locations, production trends, etc., would be included here if available.)

Table 3. Chinese REE companies and outputs (circa 2023). (Data from industry reports)

Company (Group)	Main Deposits	Annual REO Output (t)	Processing (Oxide, Metal) capacity
China Northern Rare Earth (Baotou)	Bayan Obo (Inner Mongolia)	~100,000	Multiple separation plants (light RE)
China Southern Rare Earth (Ganzhou)	Xiangshan, Longnan (Jiangxi)	~20,000	Heavy RE separation (Y,Dy)
Minmetals/China Rare Earth	Chaozhou (Guangdong), Fujian clays	10,000–20,000	Mixed light/heavy; NdFeB alloys
Chinalco (Aluminum Corp)	Maoniuping (Sichuan)	~10,000	Oxides & intermediate compounds
Xiamen Tungsten / PRC Tungsten	Bastnaesite concentrates (various)	5,000–10,000	NdFeB magnet manufacturing
Shandong Weiqiao (private)	Weishan (Shandong)	~3,000 (NdFeB plant)	NdFeB magnets (~5,000 t magnet/year)

Sources: Company reports; Chinese Society of Rare Earths; Worldmetrics [35†L133-L142] .

Figure 1. (Map) Major REE deposits and mineral provinces in China. [Not shown: would depict Bayan Obo, southern clay belt, etc.]

Figure 2. (Chart) China's REO production (2000–2024). [Not shown: would plot mine output and quota effects.]

Timeline: (Above) Key policy milestones in China's REE industry (2005–2025).

Data Gaps and Uncertainties

Published Chinese data often lacks granularity: e.g. provincial or company outputs may be aggregated or confidential. Illegal mining/exports (notably in the 2000s) significantly affected actual output but are hard to quantify. Environmental releases (fluoride, ammonia) and health impact data in mining areas are still incomplete. Mineral resource figures (reserves vs. resources) are frequently revised. Forecasts must note that China's policies (quotas, tech controls) can change rapidly, affecting supply. Despite these gaps, the convergence of multiple sources (USGS, CAS, industry associations) lends confidence to the broad trends cited here.

Sources: Chinese Ministry of Natural Resources reports, USGS Mineral Yearbooks, Chinese Society of Rare Earths publications, recent peer-reviewed studies and industry analyses (e.g. 【9†L38-L47】 【33†L277-L284】 【47†L39-L48】), and news from Chinese official media. All quantitative data are from open sources with citations as above.
